

Ratio Estimation

Sampling Theory and Methods of Estimation

Bijay Lal Pradhan, PhD

Associate Professor

Department of Statistics

Amrit Campus, Tribhuvan University, Kathmandu, Nepal

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Presentation Outline

Introduction

Concept and Motivation

Notation

Ratio Estimators

Bias and Variance

Derivation

Worked Example

Comparison with SRS

Stratified Sampling

Worked Stratified Example

Estimation Procedure

Exercises

Key Takeaways

Introduction

Why Use Auxiliary Information?

Basic Idea

In sampling theory, information on an auxiliary variable related to the study variable can be used to improve the precision of an estimator.

- ▶ Under SRS, the sample mean is a natural estimator of the population mean.
- ▶ If an auxiliary variable is strongly related to the study variable, additional information can be exploited.
- ▶ Ratio and regression estimators are important methods based on auxiliary information.
- ▶ The population value of the auxiliary variable is assumed to be known or available at low cost.

Key Principle

The stronger the relationship between y and x , the greater the potential gain from using the auxiliary information.

From SRS to Ratio Estimation

Simple Random Sampling



Sample mean $\bar{y}$



Auxiliary information becomes available



Ratio Estimation

Main Idea

If y_i/x_i is relatively stable across population units, the known population information on x can be used to estimate the unknown characteristic Y .

Concept and Motivation

What is Ratio Estimation?

Definition

The ratio method of estimation uses an auxiliary variable x_i , correlated with the study variable y_i , to obtain an improved estimate of a population total or mean.

- ▶ y_i : study variable of interest.
- ▶ x_i : auxiliary variable.
- ▶ X : known population total of the auxiliary variable.
- ▶ $\bar{X}$: known population mean of the auxiliary variable.
- ▶ The sample provides observations on both x_i and y_i .

Important Condition

The auxiliary variable should generally have a strong positive relationship with the study variable, so that the ratio y_i/x_i does not vary greatly among units.

A Practical Example: Apple Juice

Problem

A juice company wants to estimate the amount of juice obtained from a large shipment of apples before extraction.

- ▶ The total number of apples may be difficult to count.
- ▶ A sample of apples can be selected.
- ▶ Juice content y_i is measured in the sample.

Auxiliary Information

The total weight of the shipment is easy to obtain.

- ▶ x_i = weight of an apple.
- ▶ X = total weight of the shipment.
- ▶ Apple weight and juice amount are expected to be positively related.

Ratio Principle

Use the sample relationship between juice amount and apple weight to estimate total juice for the entire shipment.

When is Ratio Estimation Useful?

- ▶ The population total or mean of the auxiliary variable x is known.
- ▶ x is measured for each sampled unit.
- ▶ x and y have a strong positive relationship.
- ▶ The ratio y/x is reasonably stable across the population.
- ▶ The regression relationship is approximately through the origin, or the ratio relationship is scientifically meaningful.

Examples of Auxiliary Variables

Study variable y	Auxiliary variable x
Household expenditure	Household income
Crop production	Cultivated area
Sales revenue	Number of customers
Tree volume	Tree diameter
Juice content	Apple weight

Notation

Notation and Terminology

x_i	Auxiliary variable for the i th sampled unit
y_i	Study variable for the i th sampled unit
$x = \sum_{i=1}^n x_i$	Sample total of x
$y = \sum_{i=1}^n y_i$	Sample total of y
$\bar{x}$	Sample mean of x
$\bar{y}$	Sample mean of y
$X = \sum_{i=1}^N X_i$	Population total of X
$\bar{X} = X/N$	Population mean of X
$Y = \sum_{i=1}^N Y_i$	Population total of Y
$\bar{Y} = Y/N$	Population mean of Y
$R = Y/X = \bar{Y}/\bar{X}$	Population ratio
$\hat{R} = \bar{y}/\bar{x} = y/x$	Sample ratio estimate

Ratio Estimators

Ratio Estimate of Population Ratio

The population ratio is

$$R = \frac{Y}{X} = \frac{\bar{Y}}{\bar{X}}.$$

The sample ratio estimator is

$$\hat{R} = \frac{y}{x} = \frac{\bar{y}}{\bar{x}}.$$

Interpretation

$\hat{R}$ represents the estimated amount of y per unit of the auxiliary variable x .

Example

If y is annual sales and x is number of customers, $\hat{R}$ estimates average sales per customer based on the sample.

Ratio Estimator of Population Total

Suppose the population total X is known.

The ratio estimate of the population total Y is

$$\hat{Y}_R = \hat{R}X$$

or equivalently,

$$\hat{Y}_R = \frac{y}{x}X = \frac{\bar{y}}{\bar{x}}X.$$

Logic

Estimated ratio $\times$ known auxiliary total

$$\hat{R} \times X = \frac{\text{sample } y}{\text{sample } x} \times \text{population } X.$$

Ratio Estimator of Population Mean

Since

$$\bar{Y} = \frac{Y}{N},$$

the ratio estimate of the population mean is

$$\hat{\bar{Y}}_R = \hat{R} \bar{X}$$

or

$$\hat{\bar{Y}}_R = \frac{\bar{y}}{\bar{x}} \bar{X}.$$

If X is known, then $\bar{X} = X/N$ is known when N is known.

Remember

Total estimation uses X ; mean estimation uses $\bar{X}$.

Summary of the Three Ratio Estimates

$$\hat{R} = \frac{\bar{y}}{\bar{x}}$$

$$\hat{Y}_R = \frac{\bar{y}}{\bar{x}} X$$

$$\hat{\bar{Y}}_R = \frac{\bar{y}}{\bar{x}} \bar{X}$$

Quantity	Estimator	Known auxiliary quantity
Population ratio R	$\hat{R} = \bar{y}/\bar{x}$	X or $\bar{X}$
Population total Y	$\hat{Y}_R = \hat{R}X$	X
Population mean $\bar{Y}$	$\hat{\bar{Y}}_R = \hat{R}\bar{X}$	$\bar{X}$

Bias and Variance

Is the Ratio Estimator Unbiased?

Important Point

The ratio estimator is generally a **biased estimator** in finite samples because it is a ratio of random quantities.

However, under standard large-sample approximations,

$$E(\hat{R}) \approx R.$$

Thus, ratio estimation is often described as approximately unbiased for sufficiently large samples under appropriate conditions.

Practical Message

A small amount of bias can be acceptable when the reduction in variance is sufficiently large.

Residual Variable Behind Ratio Estimation

Define

$$u_i = y_i - Rx_i.$$

Then

$$\bar{u} = \bar{y} - R\bar{x}.$$

Since

$$R = \frac{\bar{Y}}{\bar{X}},$$

we have

$$\bar{U} = \bar{Y} - R\bar{X} = 0.$$

Interpretation

The residual $u_i = y_i - Rx_i$ measures how far the observed y_i is from the value predicted by the population ratio R times x_i .

Approximate Variance of the Ratio Estimator

Under SRS without replacement, with sampling fraction

$$f = \frac{n}{N},$$

the approximate variance of $\hat{R}$ is

$$V(\hat{R}) \approx \frac{1-f}{n\bar{X}^2} S_{y-Rx}^2$$

where

$$S_{y-Rx}^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i - RX_i)^2.$$

Equivalently,

$$V(\hat{R}) \approx \frac{1-f}{n\bar{X}^2} \frac{\sum_{i=1}^N (Y_i - RX_i)^2}{N-1}.$$

Variance of Ratio Estimates of Total and Mean

For the population total,

$$V(\hat{Y}_R) \approx \frac{N^2(1-f)}{n} S_{y-Rx}^2.$$

For the population mean,

$$V(\hat{\bar{Y}}_R) \approx \frac{1-f}{n} S_{y-Rx}^2.$$

For the ratio,

$$V(\hat{R}) \approx \frac{1-f}{n\bar{X}^2} S_{y-Rx}^2.$$

Common Structure

All three expressions depend on the residual variation S_{y-Rx}^2 .

Variance in Terms of Coefficients of Variation

Let

$$C_x = \frac{S_x}{\bar{X}}, \quad C_y = \frac{S_y}{\bar{Y}}, \quad C_{xy} = \rho C_x C_y.$$

Then the approximate variance of the ratio estimator can be expressed as

$$V(\hat{R}) \approx \frac{1-f}{n} R^2 (C_y^2 + C_x^2 - 2C_{xy})$$

and since $C_{xy} = \rho C_x C_y$,

$$V(\hat{R}) \approx \frac{1-f}{n} R^2 (C_y^2 + C_x^2 - 2\rho C_x C_y).$$

Variance Reduction: Why Correlation Matters

S_{y-Rx}^2 depends on the relationship between x and y .

For the coefficient-of-variation form,

$$C_y^2 + C_x^2 - 2\rho C_x C_y.$$

- ▶ When ρ is large and positive, the residual variation can be small.
- ▶ Smaller residual variation means smaller sampling variance.
- ▶ If the relationship is weak, ratio estimation may provide little improvement.

Key Insight

Ratio estimation is effective when the auxiliary variable explains a substantial part of the variation in the study variable.

Derivation

Derivation of $V(\hat{R})$: Step 1

Start with

$$\hat{R} = \frac{\bar{y}}{\bar{x}}.$$

For a sufficiently large sample,

$$E(\hat{R}) \approx R.$$

Therefore,

$$V(\hat{R}) \approx E(\hat{R} - R)^2.$$

Now,

$$\hat{R} - R = \frac{\bar{y}}{\bar{x}} - R = \frac{\bar{y} - R\bar{x}}{\bar{x}}.$$

For large samples, $\bar{x}$ is close to $\bar{X}$, giving

$$V(\hat{R}) \approx \frac{1}{\bar{X}^2} V(\bar{y} - R\bar{x}).$$

Derivation of $V(\hat{R})$: Step 2

Define

$$u_i = y_i - Rx_i.$$

Then

$$\bar{u} = \bar{y} - R\bar{x}.$$

Also,

$$\bar{U} = \bar{Y} - R\bar{X} = 0.$$

Under SRS without replacement,

$$V(\bar{u}) = \frac{1-f}{n} S_u^2.$$

where

$$S_u^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i - RX_i)^2.$$

Hence,

$$V(\hat{R}) \approx \frac{1-f}{n\bar{X}^2} S_u^2$$

Derivation: Final Result

Substituting

$$u_i = y_i - Rx_i$$

into the variance expression gives

$$V(\hat{R}) \approx \frac{1-f}{n\bar{X}^2} \sum_{i=1}^N \frac{(Y_i - RX_i)^2}{N-1}.$$

Therefore,

$$V(\hat{R}) = \frac{1-f}{n\bar{X}^2} \sum_{i=1}^N \frac{(Y_i - RX_i)^2}{N-1}$$

under the usual first-order approximation.

Interpretation

The variance is governed by the variability of the residuals $Y_i - RX_i$.

Worked Example

Worked Example: Estimating a Population Total

A population contains $N = 1000$ households. The known total annual income is

$$X = 50,000,000.$$

A simple random sample of $n = 100$ households gives

$$\bar{x} = 48,000, \quad \bar{y} = 12,500.$$

Estimate the population total expenditure Y using the ratio estimator.

Step 1: Estimate the ratio

$$\hat{R} = \frac{\bar{y}}{\bar{x}} = \frac{12,500}{48,000}.$$

Worked Example: Calculation

From the previous slide,

$$\hat{R} = 0.2604167.$$

The ratio estimate of total expenditure is

$$\hat{Y}_R = \hat{R}X.$$

Therefore,

$$\hat{Y}_R = 0.2604167(50,000,000)$$

$$\hat{Y}_R \approx 13,020,833.33$$

Interpretation

The estimated population total annual expenditure is approximately 13.02 million, based on the sample expenditure-to-income ratio and the known population income total.

Worked Example: Estimating the Population Mean

Suppose the known population mean income is

$$\bar{X} = 50,000.$$

Using

$$\hat{Y}_R = \hat{R}\bar{X},$$

we obtain

$$\hat{Y}_R = 0.2604167(50,000)$$

$$\hat{Y}_R \approx 13,020.83.$$

Consistency Check

Because $X = N\bar{X}$,

$$\hat{Y}_R = N\hat{\bar{Y}}_R.$$

For $N = 1000$, this gives approximately 13,020,833.33.

Worked Example: Direct Computation from Sample Totals

Suppose the sample totals are

$$x = 4,800,000, \quad y = 1,250,000.$$

Then

$$\hat{R} = \frac{y}{x} = \frac{1,250,000}{4,800,000} = 0.2604167.$$

With known population total $X = 50,000,000$,

$$\hat{Y}_R = \frac{y}{x}X.$$

$$\hat{Y}_R \approx 13,020,833.33.$$

Useful Formula

For total estimation, you can work directly with sample totals y and x .

Comparison with SRS

Ratio Estimation versus the Sample Mean

Feature	SRS estimator	Ratio estimator
Information used	Study variable only	Study + auxiliary variable
Mean estimator	$\bar{y}$	$\hat{R}\bar{X}$
Total estimator	$N\bar{y}$	$\hat{R}X$
Auxiliary total required	No	Yes
Relationship required	Not necessary	Strong x–y relationship is desirable
Main advantage	Simple	Can reduce sampling variance
Main issue	May be less precise	Can have ratio-estimation bias

Ratio Estimation: Advantages

- ▶ Uses known auxiliary information.
- ▶ Can substantially improve precision when x and y are strongly related.
- ▶ Easy to interpret through the ratio y/x .
- ▶ Useful when population auxiliary totals are available from administrative records or previous censuses.
- ▶ Particularly useful when the relationship between y and x is approximately proportional.

Ratio Estimation: Limitations

- ▶ The estimator is generally biased in finite samples.
- ▶ A weak relationship between x and y may produce little gain.
- ▶ The method is sensitive to values of x close to zero.
- ▶ The assumed ratio relationship may not be appropriate for every population.
- ▶ The auxiliary population total or mean must be known accurately.

Decision Rule

Do not use ratio estimation simply because an auxiliary variable exists. Examine whether it has a meaningful and sufficiently strong relationship with the study variable.

Stratified Sampling

Ratio Estimation in Stratified Sampling

Suppose the population is divided into L strata.

For stratum h :

N_h = population size, n_h = sample size.

Let

X_h = known auxiliary total in stratum h .

The separate ratio estimate of the population total is

$$\hat{Y}_{RS} = \sum_{h=1}^L \hat{R}_h X_h$$

where

$$\hat{R}_h = \frac{y_h}{x_h} = \frac{\bar{y}_h}{\bar{x}_h}.$$

Separate Ratio Estimator

If an independent SRS is drawn from each stratum,

$$\hat{R}_h = \frac{y_h}{x_h}.$$

The separate ratio estimator is

$$\hat{Y}_{RS} = \sum_{h=1}^L \frac{y_h}{x_h} X_h$$

For the population mean,

$$\hat{\bar{Y}}_{RS} = \sum_{h=1}^L W_h \hat{R}_h \bar{X}_h$$

where

$$W_h = \frac{N_h}{N}.$$

Combined Ratio Estimator

Instead of calculating a separate ratio in each stratum, a combined ratio can be formed using all sampled observations.

The combined sample ratio is

$$\hat{R}_c = \frac{\sum_{h=1}^L y_h}{\sum_{h=1}^L x_h}$$

The combined ratio estimate of the population total is

$$\hat{Y}_{RC} = \hat{R}_c X$$

or

$$\hat{Y}_{RC} = \frac{\sum_h y_h}{\sum_h x_h} X.$$

Separate versus Combined Ratio Estimation

	Separate	Combined
Ratio	$\hat{R}_h = y_h/x_h$ for each stratum	$\hat{R}_c = \sum y_h / \sum x_h$
Total estimate	$\sum_h \hat{R}_h X_h$	$\hat{R}_c X$
Stratum-specific relationship	Explicitly used	Pooled
Main idea	Each stratum has its own ratio	One ratio for the whole sample

Teaching Point

Separate and combined estimators use the same auxiliary-variable principle, but differ in how the ratio information is pooled across strata.

Worked Stratified Example

Stratified Example: Data

A population has two strata. The population auxiliary totals are known.

Stratum	X_h	x_h	y_h
1	6000	3000	1200
2	4000	2500	1250
Total	10000	5500	2450

Find:

1. the separate ratio estimate;
2. the combined ratio estimate.

Stratified Example: Separate Ratio

For stratum 1,

$$\hat{R}_1 = \frac{1200}{3000} = 0.40.$$

For stratum 2,

$$\hat{R}_2 = \frac{1250}{2500} = 0.50.$$

Therefore,

$$\begin{aligned}\hat{Y}_{RS} &= \hat{R}_1 X_1 + \hat{R}_2 X_2. \\ &= (0.40)(6000) + (0.50)(4000) \\ &= 2400 + 2000.\end{aligned}$$

$$\boxed{\hat{Y}_{RS} = 4400}$$

Stratified Example: Combined Ratio

The combined sample ratio is

$$\hat{R}_c = \frac{1200 + 1250}{3000 + 2500} = \frac{2450}{5500} = 0.44545.$$

Since $X = 6000 + 4000 = 10000$,

$$\hat{Y}_{RC} = \hat{R}_c X.$$

$$\hat{Y}_{RC} = (0.44545)(10000).$$

$$\hat{Y}_{RC} \approx 4454.55$$

Observation

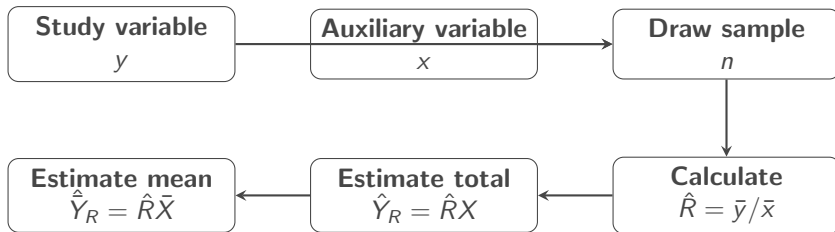
The separate and combined ratio estimators need not give the same estimate.

Estimation Procedure

General Procedure for Ratio Estimation

1. Identify the study variable y .
2. Identify a suitable auxiliary variable x .
3. Obtain the known population total X or mean $\bar{X}$.
4. Draw an SRS or stratified random sample.
5. Calculate $\bar{y}$ and $\bar{x}$, or sample totals y and x .
6. Calculate the sample ratio $\hat{R} = y/x$.
7. Estimate the population total using $\hat{Y}_R = \hat{R}X$.
8. Estimate the population mean using $\hat{\bar{Y}}_R = \hat{R}\bar{X}$.
9. If required, estimate the variance and standard error.

Ratio Estimation: A Visual Workflow



Core Calculation

Estimate the sample ratio first; then multiply by the known population auxiliary quantity.

Exercises

Practice Exercise 1: Population Total

A population has $N = 500$ units. The population total of the auxiliary variable is known to be

$$X = 25,000.$$

An SRS of $n = 50$ units gives

$$\bar{x} = 40, \quad \bar{y} = 18.$$

Find:

1. $\hat{R}$;
2. $\hat{Y}_R$;
3. $\hat{\hat{Y}}_R$.

Question: What does the estimated ratio represent in this context?

Practice Exercise 2: Comparing Estimators

Suppose the known population mean of x is $\bar{X} = 60$.

From an SRS,

$$\bar{y} = 25, \quad \bar{x} = 50.$$

Calculate:

$$\hat{\bar{Y}}_R = \frac{\bar{y}}{\bar{x}} \bar{X}.$$

Then compare it conceptually with the ordinary SRS estimator

$$\hat{\bar{Y}}_{SRS} = \bar{y}.$$

Discussion

Under what circumstances would you expect the ratio estimator to be more precise than the sample mean estimator?

Practice Exercise 3: Stratified Ratio Estimation

For three strata, the following information is available:

Stratum	X_h	x_h	y_h
1	5000	2500	900
2	3000	1800	720
3	2000	1000	350

Calculate:

1. the separate ratios $\hat{R}_1, \hat{R}_2, \hat{R}_3$;
2. the separate ratio estimate $\hat{Y}_{RS}$;
3. the combined ratio $\hat{R}_c$;
4. the combined ratio estimate $\hat{Y}_{RC}$.

Key Takeaways

Key Takeaways

1. Ratio estimation uses auxiliary information to improve estimation.
2. The auxiliary variable x should be strongly related to the study variable y .
3. The basic sample ratio is

$$\hat{R} = \frac{\bar{y}}{\bar{x}} = \frac{y}{x}.$$

4. The population total estimator is

$$\hat{Y}_R = \hat{R}X.$$

5. The population mean estimator is

$$\hat{\bar{Y}}_R = \hat{R}\bar{X}.$$

6. Ratio estimators are generally biased but can have substantially lower variance.
7. In stratified sampling, separate and combined ratio estimators can be used.

The Central Idea

Known auxiliary information



Strong relationship between x and y



Stable sample ratio



Improved estimation precision

Remember

$$\hat{Y}_R = \frac{\bar{y}}{\bar{x}} X$$

$$\hat{\hat{Y}}_R = \frac{\bar{y}}{\bar{x}} \bar{X}$$

Thank You

Questions and Discussion

Use information wisely to improve statistical estimation.

Bijay Lal Pradhan, PhD
Associate Professor, Department of Statistics
Amrit Campus, Tribhuvan University
<https://bijaylalpradhan.com.np>